# Homework 3

Module 3 · Homework · No date

Name _____ Date _____

An antibiotic with a clearance of 5.775 L/hr and an apparent volume of distribution of approximately 25 L is to be administered by IV infusion to attain a steady-state plasma concentration of 20 mcg/mL.

**a** Module 3

What is an appropriate rate of infusion to achieve the desired steady-state concentration of 20 mcg/mL?

**b** Module 3

What is the half-life of the drug that is to be infused?

**c** Module 3

What is the expected concentration of drug in the plasma 6 hours after the start of the infusion?

**d** Module 3

How much time is required to reach 95% of the steady-state concentration if no loading dose were used?

**e** Module 3

Recommend a loading dose to obtain the steady-state concentration immediately.

**f** Module 3

If the infusion were stopped after the desired steady-state concentration of 20 mcg/mL was attained, what would the concentration be 8 hours after the cessation?

**g** Module 3

If the infusion rate that you recommended in a (above) were doubled, what would the new steady-state concentration be?

**h** Module 3

If the infusion rate that you recommended in a (above) were doubled, how long would it take to reach 95% of the new steady-state concentration (no loading dose)?

**i** Module 3

If the infusion were stopped after proceeding long enough at the rate determined in part g (above) to reach the new steady-state concentration, what would the concentration be 6 hours after the cessation?

**j** Module 3

What specific change would you make to achieve a steady state concentration of 12 mg/L?